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For the Best Global Pharmaceutical Company
KOREA UNITED PHARM. INC.

KUPZINE [ku:pzi:n]

A Safer Contrast Agent that delivers Sharper MRI Images

- Indication & Administration
- Macrocyclic GBCAs: The Safer Choice
- Sharper Images & Clearer Diagnosis
- A Broadly Indicated GBCA
- Key-Messages

INDICATION & ADMINISTRATION

COMPOSITION

- Gadobutrol Monohydrate 622.73mg(as Gadobutrol 604.72mg) in 1mL of solution for injection

INDICATION

- Contrast enhancement in cranial and spinal magnetic resonance imaging (MRI).
- Contrast enhancement in magnetic resonance angiography (CE-MRA).
- Contrast enhanced MRI of liver or kidneys in patients with high suspicion or evidence of having focal lesions to classify these lesions as benign or malignant.
- Gadobutrol can also be used for MR Imaging of pathologies of the whole body.

ADMINISTRATION

[Whole Body MRI (except MRA)]

- In general, the administration of 0.1 mL(0.01 mmol)/kg is sufficient to answer the clinical question.

[CE-MRA]

- 1 FOV: <75 kg → 7.5 mL / ≥75 kg → 10 mL (0.1–0.15 mmol/kg)
- 1 FOV: <75 kg → 15 mL / ≥75 kg → 20 mL (0.2–0.3 mmol/kg)

STORAGE

- Preserve in hermetic containers. Store at room temperature not exceeding 30˚C.

WHAT IS MRI CONTRAST AGENT?

MRI Mechanism

Magnetic field

- Hydrogen protons align
- Radiofrequency(RF) pulse excites protons
- Protons relax(**T_1 = Recovery time**)
- Energy is released as RF signals
- Signal is detected(**short T_1 = strong Signal**)
- Image is formed(**strong Signal = Bright**)

MRI Contrast Agent

GBCA(Gadolinium-Based Contrast Agent) **injection**

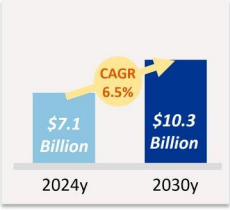
- Shortens T_1 relaxation
- Increases signal intensity

Without With GADOVA

Lesion appear brighter
Improves diagnostic clarity

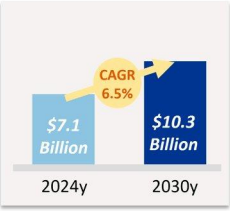
GLOBAL MARKET STATUS

MRI



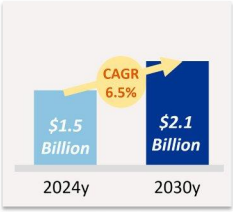
Source: RESEARCH AND MARKETS 2025

MRI Contrast Agents



Source: RESEARCH AND MARKETS 2023

Gadobutrol



Source: VERIFIED MARKET REPORTS 2025

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INDICATION & ADMINISTRATION

Macrocyclic GBCAs are the First Choice in Kidney Disease

- Macrocyclic GBCAs: Gadolinium is tightly bound within a ring structure, resulting in high stability.
- Macrocyclic GBCAs have **the fewest cases of NSF*** in patients with impaired kidney function.

* **NSF: N**ephrogenic **S**ystemic **F**ibrosis

Structure	Type	Agent	Risk Group	NSF Risk
Cyclic (Macrocyclic)	Nonionic	**Gadobutrol**	Group II	Low
		Gadoteridol		
	Ionic	Gadoterate		
Linear	Nonionic	Gadodiamide	Group I	High
		Gadoversetamide		
	Ionic	Gadopentetate		
		Gadobenate	Group II	Low
		Gadofosveset	Group III	Uncertain
		Gadoxetate		

Source: Cleveland Clinic Journal of Medicine November 2020, 87 (11): 683-694

Macrocyclic GBCAs are the First Choice in Kidney Disease

- Macrocyclic GBCAs were below the quantification limit, while linear agents showed deposition.
- Method: 3 sheep per agent received a single 0.1 mmol/kg dose(at the same clinical dose). Gadolinium concentrations in the dentate nucleus were quantified 10 weeks after injection.

Category	Agent	Gadolinium concentration (ng/g tissue)	Mean (ng/g tissue)
Control	Saline	< 5.7(below the quantification Limit)	
Macrocyclic	**Gadobutrol**	**< 5.7(below the quantification Limit)**	
	Gadoteridol		
	Gadoterate		
Linear	Gadobenate	58, 76, 116	83
	Gadodiamide	145, 154, 167	155

Source: Investigative Radiology, 2019; 54(9): 531–536

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SHARPER IMAGES & CLEARER DIAGNOSIS

MRI GBCA with the Highest Image Clarity

- Gadobutrol is a **high-concentration 1.0 M GBCA**, twice that of most other GBCAs(0.5 M)
- Among Macrocyclic GBCAs, **Gadobutrol shows the highest T_1 relaxivity.**
- Method: Plasma at 1.5 T, 3 T, and 7 T (at body temperature, 37 °C) and Whole blood at 3 T. T_1 times were measured using the inversion recovery turbo spin-echo sequence.

Contrast Agent	Plasma 1.5 T	Plasma 3 T	Plasma 7 T	Blood 3 T
Gadobutrol	**4.78**	**4.97**	**3.83**	**3.47**
Gadoteridol	3.8	3.28	3.21	2.61
Gadoterate	3.32	3	2.84	2.72

Source: Investigative Radiology 54(9):p 559-564, September 2019.

Shorter T_1 = Faster relaxation of hydrogen proton → Stronger signal intensity → Higher contrast and Sharper MRI images

A BROADLY INDICATED GBCA

The GBCA with the Broadest FDA-Approved Indications

- Among GBCA, Gadobutrol has the **Most FDA-approved Indications.**
- Gadobutrol is the **only GBCA** with an FDA-approved Indication for **Breast and Cardiac MRI.**

Agent	Body	Liver	Breast	CNS	Head&Neck	CE-MRA	Cardiac
Gadobutrol			**O**	**O**		**O**	**O**
Gadoteridol				O	O		
Gadoterate				O			
Gadodiamide	O			O			
Gadoversetamide		O		O	O		
Gadopentetate	O			O	O		
Gadobenate				O		O	
Gadofosveset						O	
Gadoxetate		O					

Source: U.S. FOOD & DRUG ADMINISTRATION, Lables for each Products

- Gadobutrol is approved for MRI of the **liver and whole body** in the United Kingdom.

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KEY-MESSAGES

01	By 2030, the estimated Global Market Size of MRI Contrast Agent is **U$3.4 Bn**, with Gadobutrol at **US$2.1Bn**.
02	Macrocyclic GBCAs demonstrates safety, having the **fewest NSF cases** and **negligible brain deposition**.
03	Among Macrocyclic GBCAs, Gadobutrol has the **highest T_1 relaxivity**, providing the **superior contrast**.
04	Gadobutrol is the **only** GBCA approved for **Breast and Cardiac MRI**, with the **broadest FDA indications**.

CONTACT KUP

STEP1 : Visit http://biz.kup.co.kr and **Press** the "Get Started" button

STEP2 : Choose the right overseas sales expert who is in charge of your region

GENERAL	ASIA, CIS, EUROPE	MIDDLE EAST, AFRICA	NORTH & SOUTH AMERICA
Mr. HANS CHO	Ms. ASPYN JEONG	Mr. SANGHUN KIM	Mr. FRANK SHIM
hanscho@kup.co.kr	jsb10652@kup.co.kr	robertkim@kup.co.kr	sangwanshim@kup.co.kr

STEP3 : Discuss whatever you want via E-mail

Omnihexol(Iohexol): The Trustworthy Choice for Clarity and Safety

1. Outstanding Safety

- Non-ionic, low-osmolar contrast agent with reduced discomfort and a lower incidence of adverse reactions.

2. Excellent Image Quality

- Delivers high-contrast resolution, enabling precise visualization of vessels, organs, and lesions.

3. Versatile Clinical applications

- Proven efficacy across a wide range of diagnostic uses, including **CT imaging, angiography, and myelography**.

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